

Technical Datasheet — VBF-T2R1FLASH-DSH-01

****Case**:** t2_r1_flash · Grid energy storage cell (virtual design, simulation-verified)

Field	Value	Source
Rated capacity	5.0 Ah nominal (parameter set); 5.0469 Ah verified @1C 25C	cell/r2_fcc_1c_dfn.json
Nominal voltage / window	midpoint 3.877 V; 2.5 – 4.2 V	calc-energy midpoint_voltage_v / parameter set
Rated energy	18.178 Wh (simulation integration $V \cdot I \cdot dt$)	cell/r2_fcc_energy.json
Energy density	425.4 Wh/kg (contract caliber, electrolyte excluded); 881.5 Wh/L	cell/r2_fcc_energy.json
Max continuous discharge rate	1C verified (5.05 Ah, 1C to 2.5 V)	cell/r2_fcc_1c_dfn.json
Fast-charge capability	4C @45C amb: no plating (anode min +12.3 mV), T_max 329.2 K with h=60 W/m2K	cell/r2_fcc_4c.json
Operating temperature range	Simulated: 25 C (1C/aging), -20 C (lowT), 45 C (4C fast charge). Full range not simulated	simulation conditions
Cycle life (SEI)	SEI 78.2 nm @100 cyc, 262.0 nm @500 cyc (1C, standard SEI model); capacity-based cycle life to 80% SOH not simulated (standard-SEI-model capacity artifact, honestly noted)	cell/r2_fcc_aging100.json / derived
Safety determination	4C fast charge: no plating, T_max below red line -> PASS	cell/r2_fcc_4c.json
Dimensions and mass	65.0 x 1580 x 0.2008 mm stack (no casing); 42.73 g (electrolyte-excluded caliber)	calc-energy
DC resistance	2.892 mOhm (10% DOD method)	cell/r2_fcc_energy.json
Power density	33.9 kW/kg ($V_{OC}^2 / 4DCR$ / mass)	cell/r2_fcc_energy.json